

रामानुजन महाविद्यालय

नैक मान्यता: ग्रेड ए ++ (३.७१)

दिल्ली विश्वविद्यालय

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MINOR PROJECT - II

BRAIN TUMUOR DETECTION SYSTEM DOCUMENTATION



B.voc Software Development

Submitted By: -

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GitHub Link: [click here !!!](#)

Live Project: [click here !!!](#)

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1.Introduction.

This project is an AI-powered Brain Tumour Detection System that leverages deep learning techniques to analyse MRI scans and detect potential brain tumors. The system provides an easy-to-use interface where users can upload MRI images, receive instant analysis, and generate a downloadable PDF report with the results.

2.Problem Statement.

Brain tumors are life-threatening conditions that require **early** and **accurate** diagnosis. Manual inspection of MRI scans is time-consuming and may be prone to human errors. This system aims to automate the detection process, reducing diagnostic time and improving accuracy.

3.Objectives.

1. Automate the detection of brain tumors using deep learning.
2. Develop a user-friendly interface for clinicians.
3. Provide reliable, real-time detection with performance metrics.
4. Generate professional diagnostic reports.

4.System Overview.

The system allows users to upload **MRI scans**, processes the images using a fine-tuned **YOLOv11m** model, highlights tumors, and generates diagnostic reports. Built with **Streamlit**, **PyTorch**, **OpenCV**, and **FPDF**.

5. Project Timeline.

Phase 1: Requirement Analysis and Data Collection	Week 1
Phase 2: Data Preparation - Cleaning, Preprocessing, and Augmentation	Week 2
Phase 3: Model Development - Fine-tuning YOLOv11 for tumor detection and testing	Week 3
Phase 4: System Design - Implementing Streamlit UI and integrating FPDF for report generation	Week 4
Phase 5: Deployment - Deploying on Streamlit Cloud and performing functional testing	Week 5
Phase 6: Final Testing and Delivery - Validating system performance, refining UI, and generating reports	Week 6

6. Technologies Used.

1. **Streamlit:**
 - For building the web interface.
2. **OpenCV:**
 - For image processing and pre-processing.
3. **NumPy:**
 - For handling numerical operations.
4. **YOLO (Ultralytics):**
 - Deep learning model for tumour detection.
5. **Pillow (PIL):**
 - For handling image formats.
6. **FPDF:**
 - For generating PDF reports.

7. Installation and Setup.

1. Clone the repository.

```
- git clone https://github.com/PresidentLivingstone/Brain_tumor_detection-system-using-finet-tuned-yolov11.git
```

2. Create a virtual environment (MacOS).

```
- python3 -m venv venv  
- source venv/bin/activate
```

3. Install dependencies.

```
- pip install -r requirements.txt
```

4. Run.

```
- Streamlit run app.py
```

8. Model Training

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Brain Tumor Detection and Localization Using YOLO v12 and Deep Learning Techniques in Python.ipynb - Colab

Brain Tumor Detection and Localization Using YOLO v12 and Deep Learning Techniques in Python

```
# check for GPU availability
!nvidia-smi
```

Tue Feb 25 11:22:36 2025

NVIDIA-SMI 550.54.15				Driver Version: 550.54.15				CUDA Version: 12.4			
GPU	Name	Perf	Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr. ECC	GPU-Util	Compute M.	ECC	M.
Fan	Temp		Pwr:Usage/Cap		Memory-Usage	GPU-Util					
0	Tesla T4		Off	00000000:00:04:0	Off						
N/A	70C	P8	11W / 70W		0MiB / 15360MiB	0%			Default		N/A

Processes:										
GPU	GI	CI	PID	Type	Process name				GPU Memory	Usage
	ID	ID								
No running processes found										

```
import torch
torch.cuda.empty_cache()
```

Step 01 # Install the Ultralytics Package

```
# install ultralytics
!pip install ultralytics
```

Show hidden output

Step 02 # Import All the Required Libraries

```
# import libraries
import ultralytics
ultralytics.checks()
from ultralytics import YOLO
from IPython.display import Image
```

Ultralytics 8.3.78 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)
Setup complete (2 CPUs, 12.7 GB RAM, 33.2/112.6 GB disk)

Step # 03 Download Dataset from Roboflow

```
# load the dataset
!pip install roboflow

from roboflow import Roboflow
rf = Roboflow(api_key="0zA9553LIS7TirJ03Cr4")
project = rf.workspace("aykut-izfoo").project("brain-tumor-detection-rwfb")
version = project.version(3)
dataset = version.download("yolov11")
```

Show hidden output

```
# check for base directory
dataset.location
```

```
'/content/Brain-Tumor-Detection-3'
```

Step # 04 Train YOLOv12 Model on a Custom Dataset

```
!yolo task=detect mode=train data={dataset.location}/data.yaml model="yolo11m.pt" epochs=20 imgsz=640 batch=8
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
12/20 7.79G 1.387 0.9343 1.569 8 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.854 0.874 0.89 0.484

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
13/20 7.79G 1.367 0.9299 1.575 8 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.873 0.83 0.877 0.507

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
14/20 7.79G 1.337 0.8876 1.517 10 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.881 0.843 0.871 0.516

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
15/20 7.74G 1.296 0.8434 1.486 13 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.863 0.843 0.892 0.53

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
16/20 7.79G 1.287 0.8211 1.487 8 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.925 0.824 0.88 0.532

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
17/20 7.79G 1.267 0.7794 1.473 8 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.911 0.874 0.921 0.552

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
18/20 7.79G 1.248 0.7439 1.46 9 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.892 0.883 0.895 0.538

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
19/20 7.74G 1.244 0.73 1.44 8 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.876 0.89 0.904 0.55

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
20/20 7.79G 1.202 0.7193 1.413 8 640: 100% 150/150 [01:53<00:00,
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.908 0.874 0.89 0.549
```

20 epochs completed in 0.666 hours.

Optimizer stripped from runs/detect/train/weights/last.pt, 40.5MB

Optimizer stripped from runs/detect/train/weights/best.pt, 40.5MB

Validating runs/detect/train/weights/best.pt...

Ultralytics 8.3.78 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)

YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs

```
Class Images Instances Box(P R mAP50 mAP50-95): 100% 10/10 [00:0.
all 150 159 0.914 0.874 0.921 0.552
```

Speed: 0.3ms preprocess, 24.9ms inference, 0.1ms loss, 2.4ms postprocess per image

Results saved to **runs/detect/train**

Learn more at <https://docs.ultralytics.com/modes/train>

Summary

- **Training:** Completed 20 epochs in **0.668 hours**.

- **Model Details:** YOLOv11m with **125 layers**, **20M parameters**, and **67.6 GFLOPs**.

- **Performance Metrics:**

- **mAP@50:** 92.1%
- **mAP@50-95:** 55.2%
- **Precision:** 91.4%
- **Recall:** 87.4%

- **Speed:**

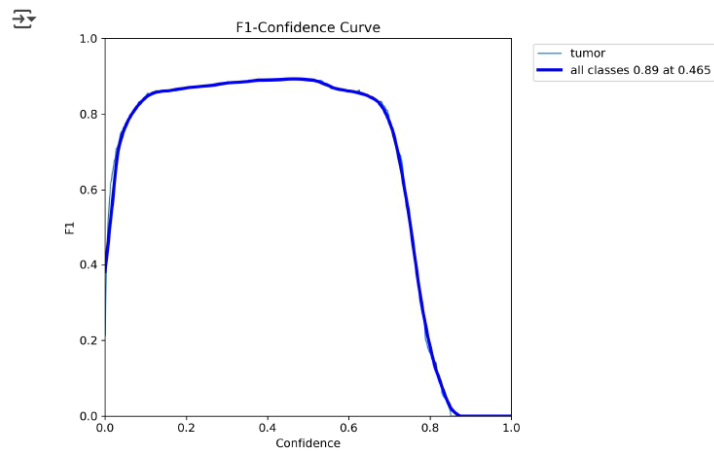
- **Inference time:** 24.9ms per image
- **Preprocessing:** 0.4ms
- **Postprocessing:** 3.4ms

- **Output Saved:** runs/detect/train/weights/

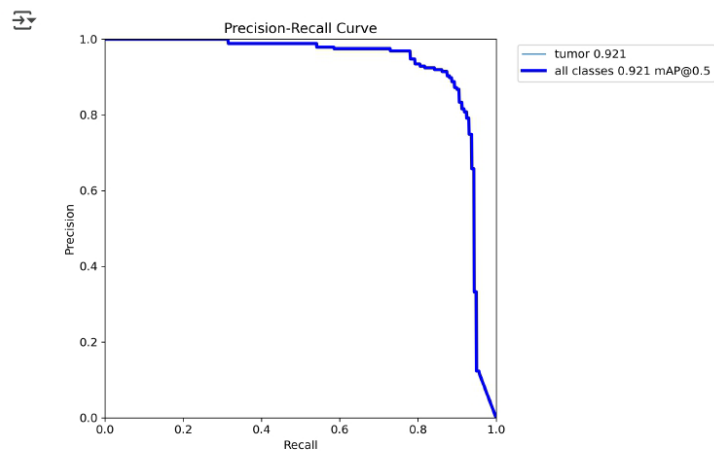
- **Optimizer Removed:** Model weights (best.pt & last.pt) stripped to 40.5MB.

✓ Step # 05 Examine Training Results

Image("/content/runs/detect/train/F1_curve.png", width=600)



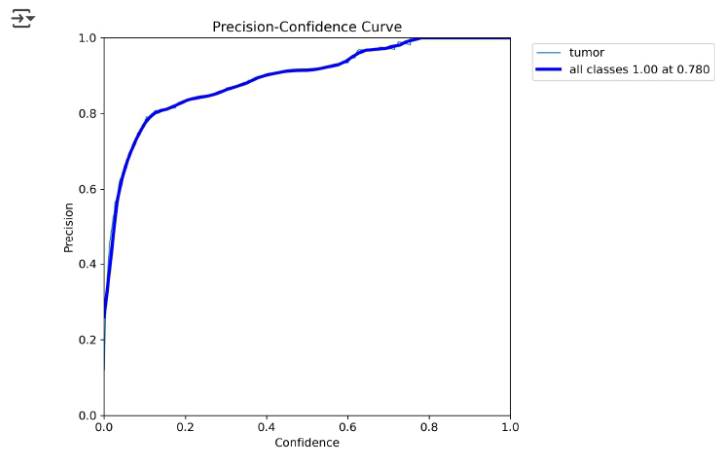
Image("/content/runs/detect/train/PR_curve.png", width=600)



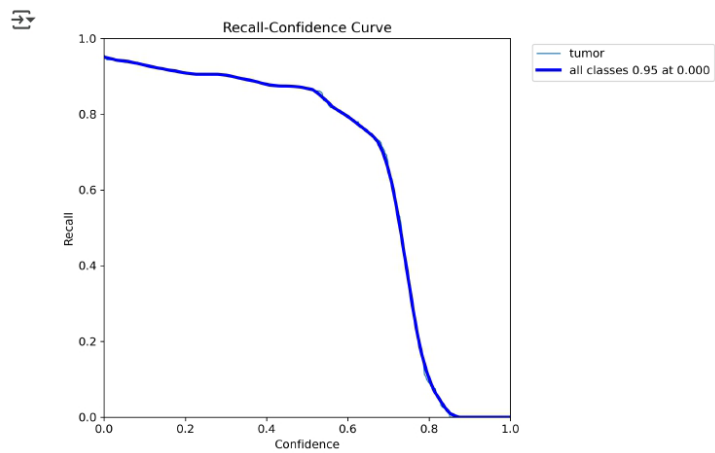
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```
Image("/content/runs/detect/train/P_curve.png", width=600)
```



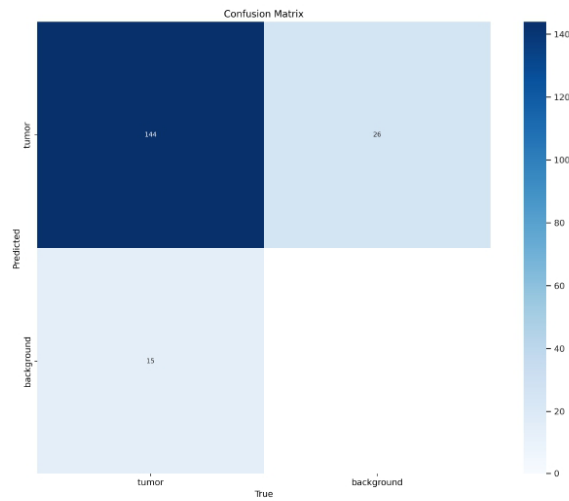
```
Image("/content/runs/detect/train/R_curve.png", width=600)
```



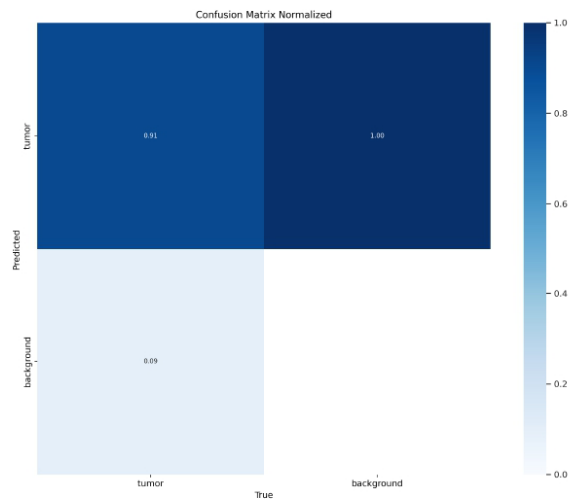
```
Image("/content/runs/detect/train/confusion_matrix.png", width=600)
```

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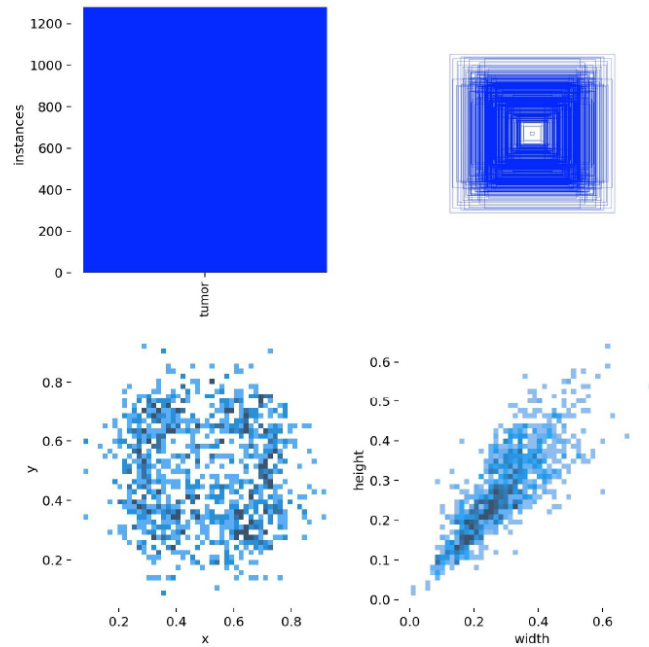
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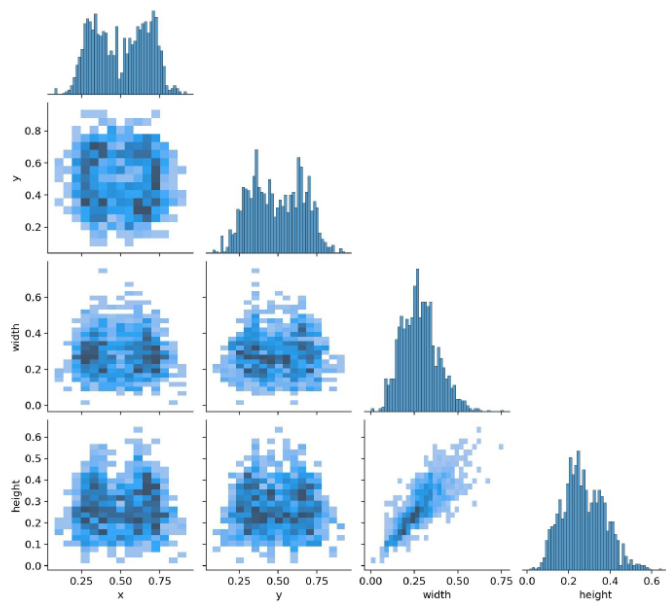
Image("/content/runs/detect/train/confusion_matrix_normalized.png", width=600)



Image("/content/runs/detect/train/labels.jpg", width=600)



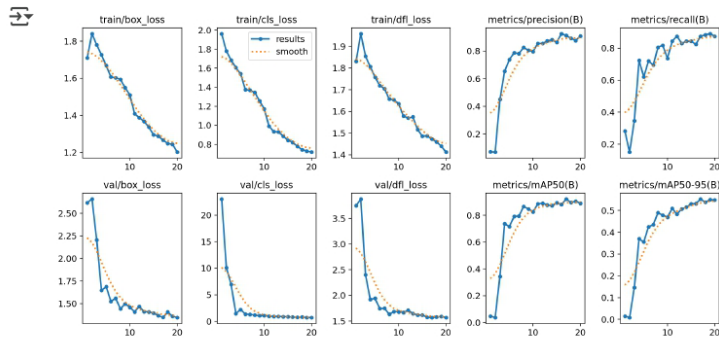
Image("/content/runs/detect/train/labels_correlogram.jpg", width=600)



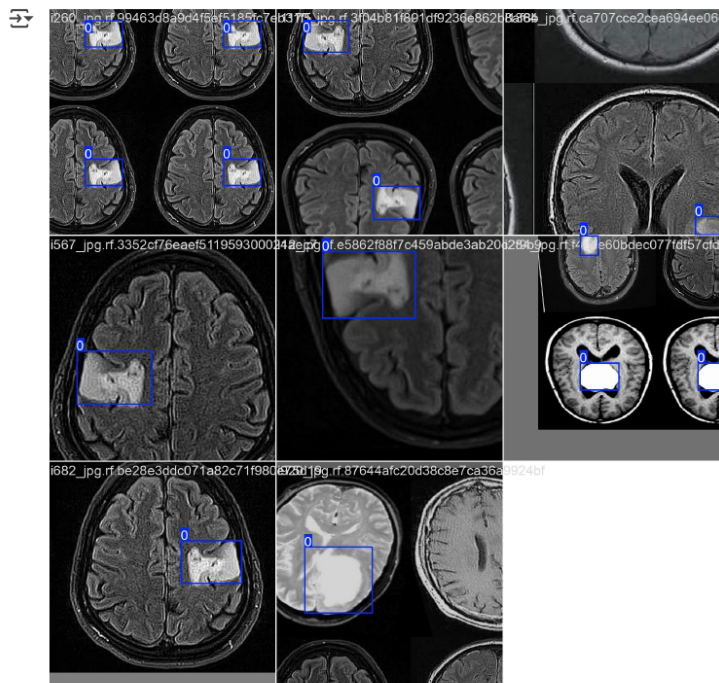
Image("/content/runs/detect/train/results.png", width=600)

27/02/2025, 18:13

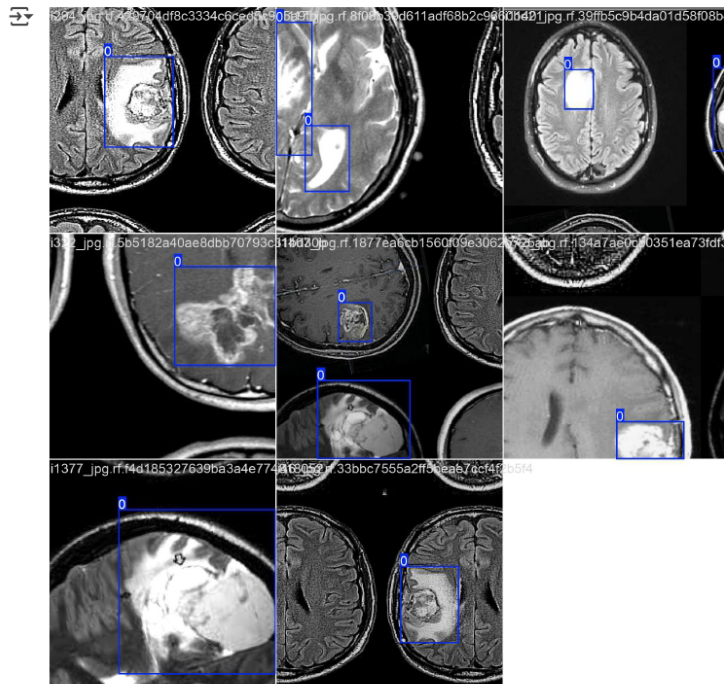
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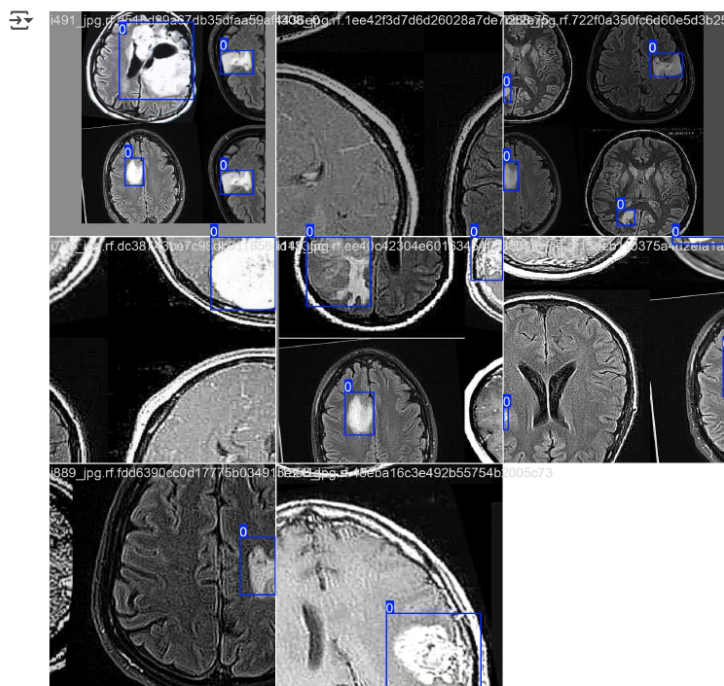
Image("/content/runs/detect/train/train_batch0.jpg", width=600)



Image("/content/runs/detect/train/train_batch1.jpg", width=600)



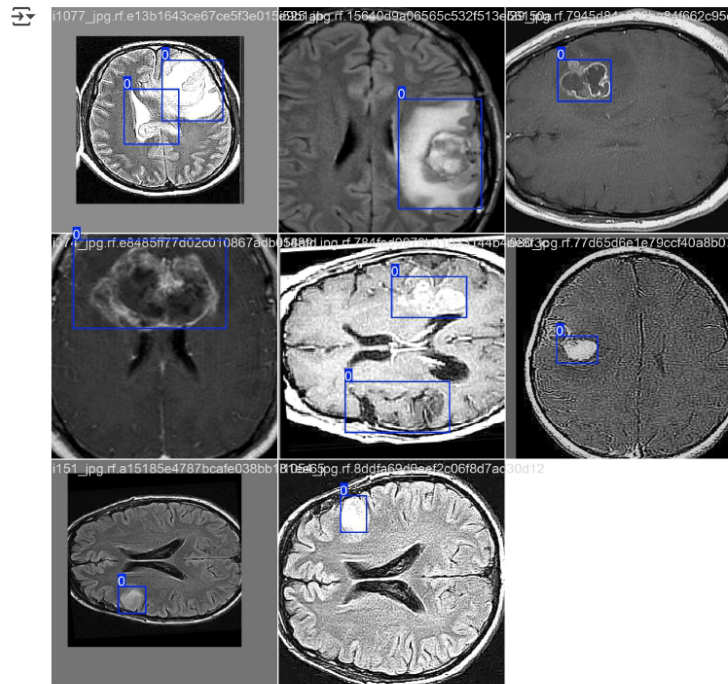
Image("/content/runs/detect/train/train_batch2.jpg", width=600)



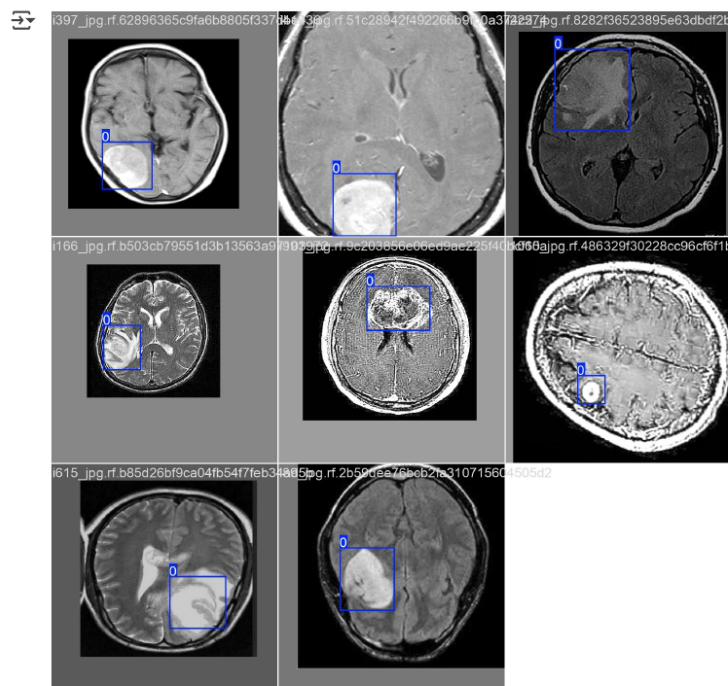
Image("/content/runs/detect/train/train_batch1500.jpg", width=600)

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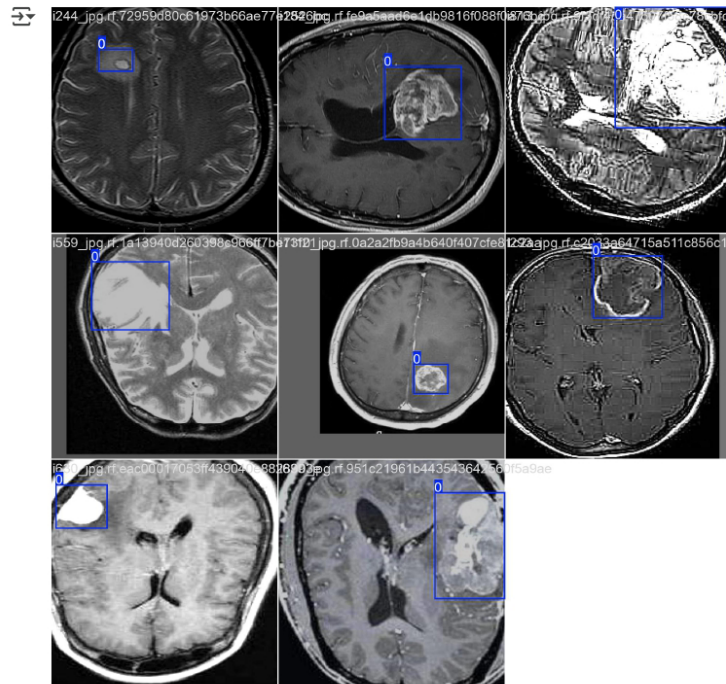
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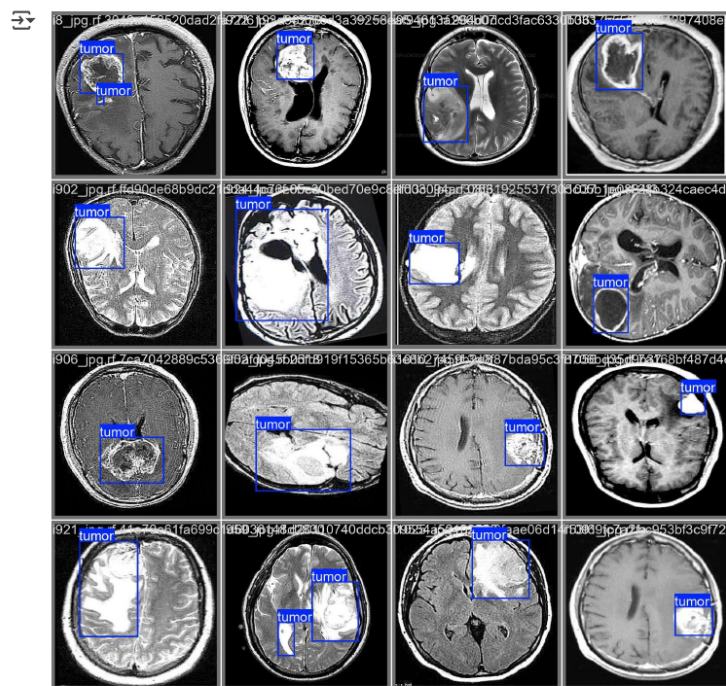
Image("/content/runs/detect/train/train_batch1501.jpg", width=600)



Image("/content/runs/detect/train/train_batch1502.jpg", width=600)



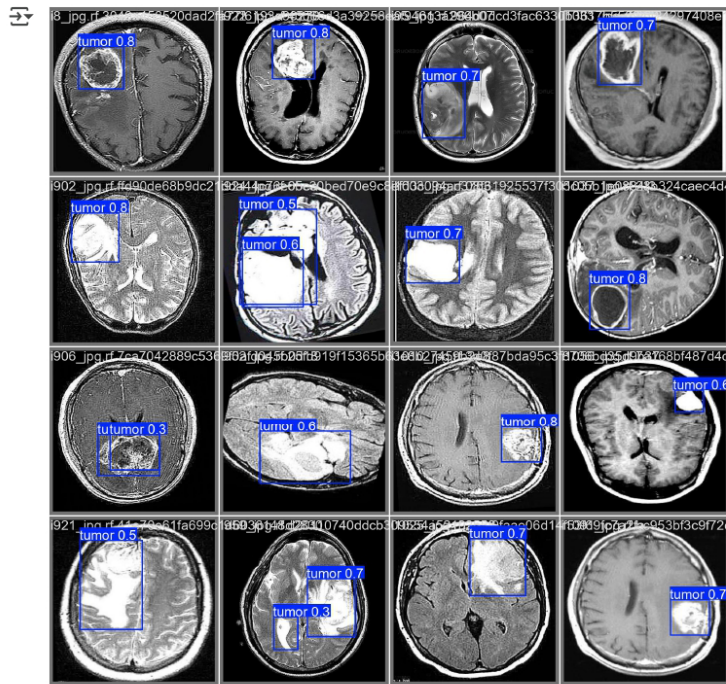
Image("/content/runs/detect/train/val_batch0_labels.jpg", width=600)



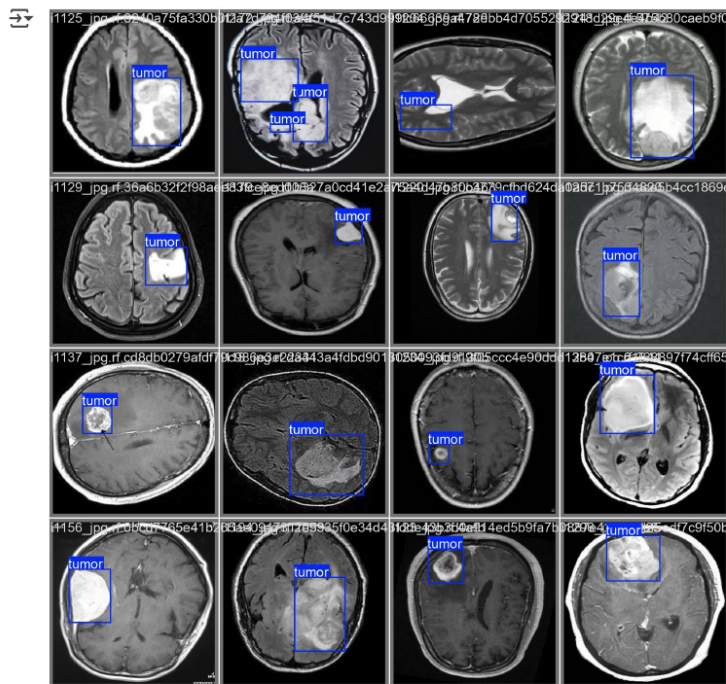
Image("/content/runs/detect/train/val_batch0_pred.jpg", width=600)

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Image("/content/runs/detect/train/val_batch1_labels.jpg", width=600)



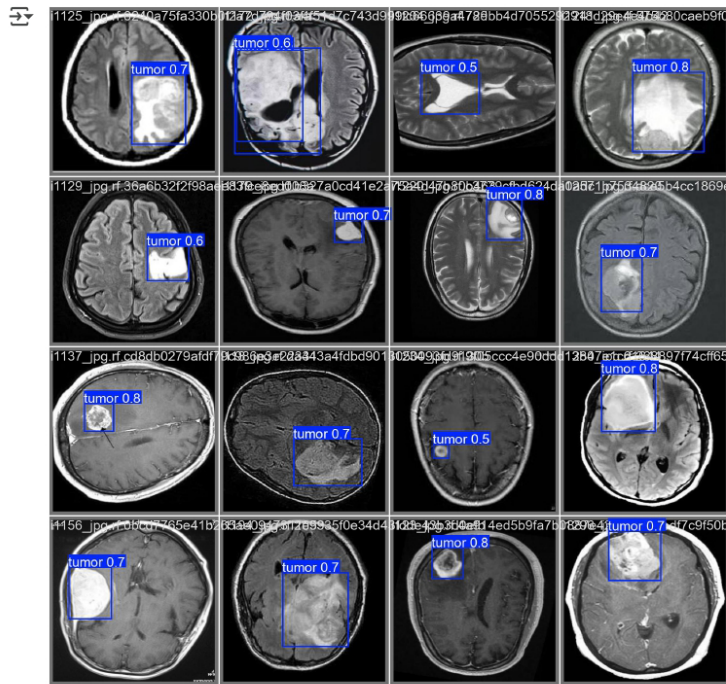
Image("/content/runs/detect/train/val_batch1_pred.jpg", width=600)

<https://colab.research.google.com/drive/1NeAQ7yrelXTiO4kfoZr-ensXIN41lek#scrollTo=odLIYyGvdSYq&printMode=true>

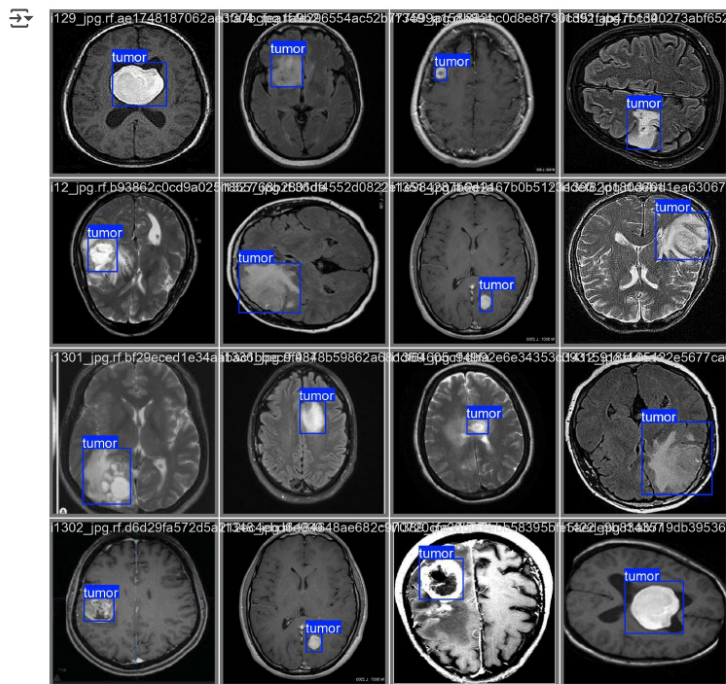
11/16

27/02/2025, 18:13

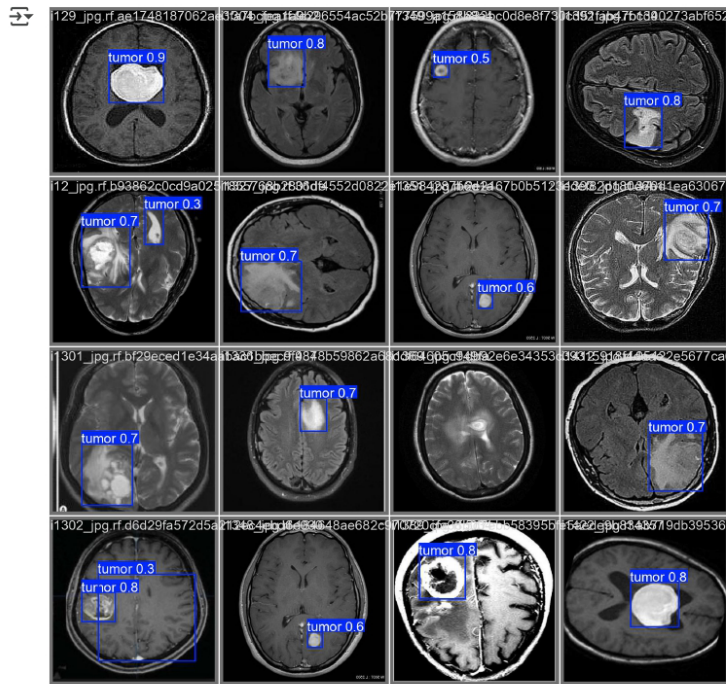
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Image("/content/runs/detect/train/val_batch2_labels.jpg", width=600)



Image("/content/runs/detect/train/val_batch2_pred.jpg", width=600)



✓ Step # 06 Validate Fine-Tuned Model

```
!yolo task=detect mode=val model="/content/runs/detect/train/weights/last.pt" data={dataset.location}/data.yaml
```

Ultralytics 8.3.78 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)
 YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs
 val: Scanning /content/Brain-Tumor-Detection-3/valid/labels.cache... 150 images, 0 backgrounds, 0 corrupt:

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95
all	150	159	0.908	0.874	0.89	0.549

 Speed: 4.2ms preprocess, 23.7ms inference, 0.0ms loss, 2.0ms postprocess per image
 Results saved to **runs/detect/val**
 Learn more at <https://docs.ultralytics.com/modes/val>

Validation Summary

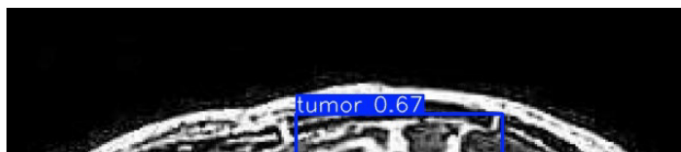
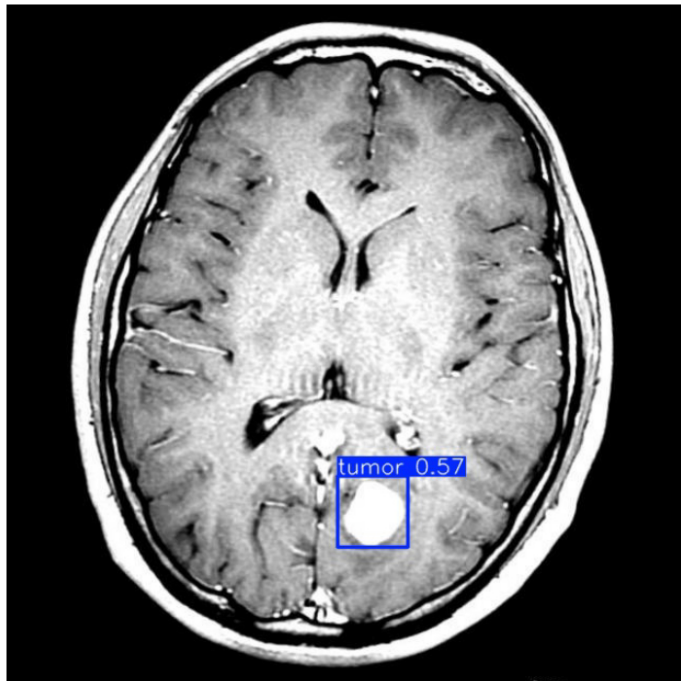
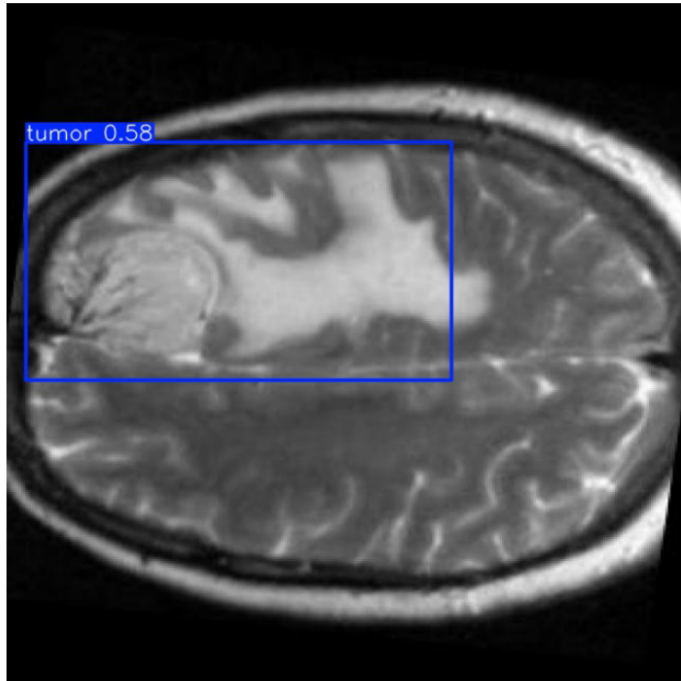
- **Model:** YOLOv11m (Ultralytics 8.3.78)
- **Hardware:** Tesla T4 (CUDA)
- **Dataset** 150 images, 159 instances, no corrupt data
- **Performance Metrics:**
 - mAP@50: 89.0%
 - mAP@50-95: 54.9%
 - Precision: 90.8%
 - Recall: 87.4%
- **Speed:**
 - Inference: 25.5ms per image
 - Preprocessing: 0.3ms
 - Postprocessing: 2.5ms
- **Results Saved:** runs/detect/val/

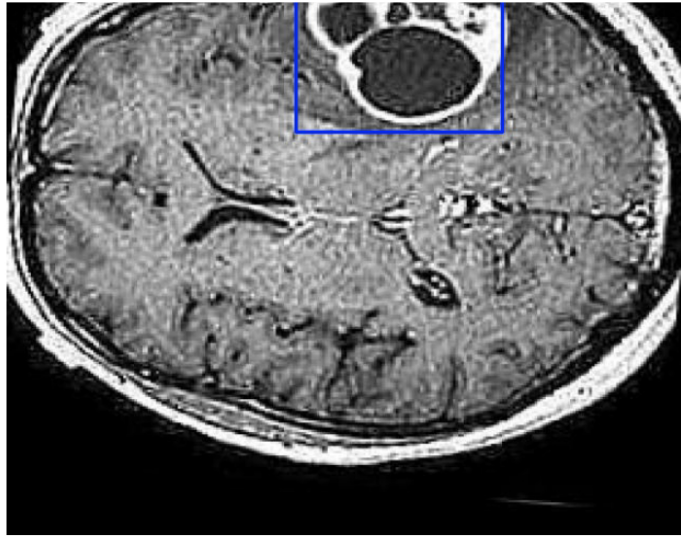
✓ Step # 07 Inference with Custom Model on Images

```
!yolo task=detect mode=predict model="/content/runs/detect/train/weights/best.pt" conf=0.25 source={dataset, lo
 image 96/150 /content/Brain-Tumor-Detection-3/test/images/i588.jpg.rf.6b3391fa86a9d4aff82a8f521391368c.jpg
image 97/150 /content/Brain-Tumor-Detection-3/test/images/i58.jpg.rf.8506a4568133f10cd507adef43982620.jpg:
image 98/150 /content/Brain-Tumor-Detection-3/test/images/i623.jpg.rf.9bdbcabcfcb5e51bd5a2724771e80d98.jpg
image 99/150 /content/Brain-Tumor-Detection-3/test/images/i632.jpg.rf.297489cc9d56e7d262e83545e977ea7a.jpg
image 100/150 /content/Brain-Tumor-Detection-3/test/images/i637.jpg.rf.46c2211bedfd6f6cedfd715855c640da.jpg:
image 101/150 /content/Brain-Tumor-Detection-3/test/images/i638.jpg.rf.76b6088d4ba38314b4372a6d2efbca3b.jpg:
image 102/150 /content/Brain-Tumor-Detection-3/test/images/i641.jpg.rf.a4954f7f47ec8613bae4a57cbce730e0.jpg:
image 103/150 /content/Brain-Tumor-Detection-3/test/images/i653.jpg.rf.e4e6062de8130ad6d2038151f592f23a.jpg:
image 104/150 /content/Brain-Tumor-Detection-3/test/images/i666.jpg.rf.23aa2b1551189b8ad68e77b03d12f545.jpg:
image 105/150 /content/Brain-Tumor-Detection-3/test/images/i669.jpg.rf.4e56b286ed4308928dbd96b50c25ef3e.jpg:
image 106/150 /content/Brain-Tumor-Detection-3/test/images/i675.jpg.rf.224e3e23da129a4a0b5b3a098dd16ec4.jpg:
image 107/150 /content/Brain-Tumor-Detection-3/test/images/i683.jpg.rf.b2dfef6ca9297f9364b618e982144f75.jpg:
image 108/150 /content/Brain-Tumor-Detection-3/test/images/i694.jpg.rf.13e27026ee003b9053d430d68152c1a.jpg:
image 109/150 /content/Brain-Tumor-Detection-3/test/images/i712.jpg.rf.567d079fcdcf99d48d17867278527d8.jpg:
image 110/150 /content/Brain-Tumor-Detection-3/test/images/i724.jpg.rf.4c51b2c0c5719b9b454f02d9829a58f8.jpg:
image 111/150 /content/Brain-Tumor-Detection-3/test/images/i726.jpg.rf.a5664c07c58928c8d2c9574ca752a33d.jpg:
image 112/150 /content/Brain-Tumor-Detection-3/test/images/i734.jpg.rf.099f749013745047d94661a66722a9a9.jpg:
image 113/150 /content/Brain-Tumor-Detection-3/test/images/i737.jpg.rf.aff5ce741a41fc19e86202e7e31f82c3.jpg:
image 114/150 /content/Brain-Tumor-Detection-3/test/images/i747.jpg.rf.5f606b9859adebf1af26ed621f4bb592.jpg:
image 115/150 /content/Brain-Tumor-Detection-3/test/images/i748.jpg.rf.b17f73f2c20648630ace8d65018b0aa1.jpg:
image 116/150 /content/Brain-Tumor-Detection-3/test/images/i758.jpg.rf.e88ebf5c54cd3ca39f347a8fee35ac87.jpg:
image 117/150 /content/Brain-Tumor-Detection-3/test/images/i761.jpg.rf.02f776ba1f0d601e560792735cdc40b0.jpg:
image 118/150 /content/Brain-Tumor-Detection-3/test/images/i771.jpg.rf.4207b274e51e172c643ea7948cfea397.jpg:
image 119/150 /content/Brain-Tumor-Detection-3/test/images/i789.jpg.rf.ea9b8127a52fb24ea03907e4fbcf4147.jpg:
image 120/150 /content/Brain-Tumor-Detection-3/test/images/i805.jpg.rf.a955bbb1b73ab245524d89ced3a0563f.jpg:
image 121/150 /content/Brain-Tumor-Detection-3/test/images/i807.jpg.rf.314992ca981c655d2eea5b7cfe2254de.jpg:
image 122/150 /content/Brain-Tumor-Detection-3/test/images/i819.jpg.rf.cc3e33a1726054de9372342b2ae4eec8.jpg:
image 123/150 /content/Brain-Tumor-Detection-3/test/images/i827.jpg.rf.324fc10d039bbebb131b342def5b873a.jpg:
image 124/150 /content/Brain-Tumor-Detection-3/test/images/i828.jpg.rf.62524a984eaa937c24d2fc5c66ed3a19.jpg:
image 125/150 /content/Brain-Tumor-Detection-3/test/images/i82.jpg.rf.b3eb7b97647849aca8c8cfaac3bdd991.jpg:
image 126/150 /content/Brain-Tumor-Detection-3/test/images/i833.jpg.rf.4e9845b2514a319c89faca07cfc68b279.jpg:
image 127/150 /content/Brain-Tumor-Detection-3/test/images/i843.jpg.rf.a59ba973eeb237c90553538e26c12472.jpg:
image 128/150 /content/Brain-Tumor-Detection-3/test/images/i846.jpg.rf.d24a8bf1cc973173c469ea8e72a4c14f.jpg:
image 129/150 /content/Brain-Tumor-Detection-3/test/images/i857.jpg.rf.02ee23f4c9397045a1c3d960b79b24b5.jpg:
image 130/150 /content/Brain-Tumor-Detection-3/test/images/i868.jpg.rf.c8a0b2fe11eb41a75b18ad55d212fd89.jpg:
image 131/150 /content/Brain-Tumor-Detection-3/test/images/i884.jpg.rf.4fcd40fe90b3926fea7294c4fffb1e61c.jpg:
image 132/150 /content/Brain-Tumor-Detection-3/test/images/i896.jpg.rf.4c5b309df255e5688366999b08e8e34e.jpg:
image 133/150 /content/Brain-Tumor-Detection-3/test/images/i908.jpg.rf.12514bd53bedd5b736af703e79f3f1be.jpg:
image 134/150 /content/Brain-Tumor-Detection-3/test/images/i910.jpg.rf.33913149831b29dd515fdb9446edb1c.jpg:
image 135/150 /content/Brain-Tumor-Detection-3/test/images/i912.jpg.rf.8f72c97302d1dbc224597f796c61a8ca.jpg:
image 136/150 /content/Brain-Tumor-Detection-3/test/images/i926.jpg.rf.861915433eedb4352757f3d0d6b2b6db.jpg:
image 137/150 /content/Brain-Tumor-Detection-3/test/images/i935.jpg.rf.95016f5f65086d3a8c3e2a62672bed1a.jpg:
image 138/150 /content/Brain-Tumor-Detection-3/test/images/i941.jpg.rf.08a84ea2cc0abaad980adb41f14bf342.jpg:
image 139/150 /content/Brain-Tumor-Detection-3/test/images/i965.jpg.rf.52bd1c05abb6cfe082cfb8c5a70b3f36.jpg:
image 140/150 /content/Brain-Tumor-Detection-3/test/images/i969.jpg.rf.c441e283f4968ef217dfd515d88b3e3e.jpg:
image 141/150 /content/Brain-Tumor-Detection-3/test/images/i96.jpg.rf.7ca0d84fc0783bba267f125e4ed1c620.jpg:
image 142/150 /content/Brain-Tumor-Detection-3/test/images/i970.jpg.rf.174ea2ccc37f0523214af069eb705615.jpg:
image 143/150 /content/Brain-Tumor-Detection-3/test/images/i983.jpg.rf.eb2a296d825cde6962d76a841d0f5d12.jpg:
image 144/150 /content/Brain-Tumor-Detection-3/test/images/i987.jpg.rf.3f5bf6959857deed8375852d6d977f2d.jpg:
image 145/150 /content/Brain-Tumor-Detection-3/test/images/i988.jpg.rf.a30953ab658ccbd589aec95987ca4650.jpg:
image 146/150 /content/Brain-Tumor-Detection-3/test/images/i98.jpg.rf.b0c828ac25c2196641290302201bcf6f.jpg:
image 147/150 /content/Brain-Tumor-Detection-3/test/images/i990.jpg.rf.4bce0a3e80e3ef872db7d8c055718634.jpg:
image 148/150 /content/Brain-Tumor-Detection-3/test/images/i992.jpg.rf.cffa34774438e2ae449765a6a680757.jpg:
image 149/150 /content/Brain-Tumor-Detection-3/test/images/i993.jpg.rf.e5ba6e0bcae9fc7be2ee42f17981eb18.jpg:
image 150/150 /content/Brain-Tumor-Detection-3/test/images/i995.jpg.rf.2fdc64de1cf961e2344ef45f642961e29.jpg:
Speed: 2.1ms preprocess, 20.3ms inference, 2.2ms postprocess per image at shape (1, 3, 640, 640)
Results saved to runs/detect/predict
🔗 Learn more at https://docs.ultralytics.com/modes/predict
```

```
import glob
import os
from IPython.display import Image as IPyImage, display

latest_folder = max(glob.glob('/content/runs/detect/predict*'), key=os.path.getmtime)
for img in glob.glob(f'{latest_folder}/*.jpg')[1:4]:
    display(IPyImage(filename=img, width=600))
    print("\n")
```





```
# convert model to torchscript
model = YOLO("/content/runs/detect/train/weights/best.pt")
model.export(format="torchscript")
```

Ultralytics 8.3.78 Python-3.11.11 torch-2.5.1+cu124 CPU (Intel Xeon 2.00GHz)
 YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs

PyTorch: starting from '/content/runs/detect/train/weights/best.pt' with input shape (1, 3, 640, 640) BCHW

TorchScript: starting export with torch 2.5.1+cu124...

TorchScript: export success 7.4s, saved as '/content/runs/detect/train/weights/best.torchscript' (77.1 MB)

Export complete (10.9s)

Results saved to /content/runs/detect/train/weights

Predict: yolo predict task=detect model=/content/runs/detect/train/weights/best.torchscript imgsz=640

Validate: yolo val task=detect model=/content/runs/detect/train/weights/best.torchscript imgsz=640

Visualize: <https://netron.app>

/content/runs/detect/train/weights/best.torchscript'

```
import cv2
import torch
import matplotlib.pyplot as plt
from ultralytics import YOLO
```

```
def predict_and_display_v11(image_path, model_path, confidence_threshold=0.5, target_size=(640, 640)):
```

```
    """
```

```
    Detects and visualizes brain tumor presence in an image using a fine-tuned YOLOv11 model.
    Enhances the image by resizing and converting it to grayscale before prediction.
```

```
    Args:
```

```
        image_path: Path to the input image.
```

```
        model_path: Path to the trained YOLOv11 TorchScript model.
```

```
        confidence_threshold: Minimum confidence score for a detection to be considered.
```

```
        target_size: The target size for resizing the image.
```

```
    """
```

```
    # Load and preprocess the image
```

```
    img = cv2.imread(image_path)
```

```
    img_resized = cv2.resize(img, target_size)
```

9. Model Description

Performance metrics:

Model Details.

- **Model:** YOLOv11m
- **Architecture:** 125 layers
- **Parameters:** 20M
- **GFLOPs:** 67.6

Performance Metrics.

- **mAP@50:** 94.2%
- **mAP@50-95:** 55.2%
- **Precision:** 92.7%
- **Recall:** 93.5%

Speed.

- **Inference Time:** 24.9ms per image
- **Preprocessing Time:** 0.4ms
- **Postprocessing Time:** 3.4ms

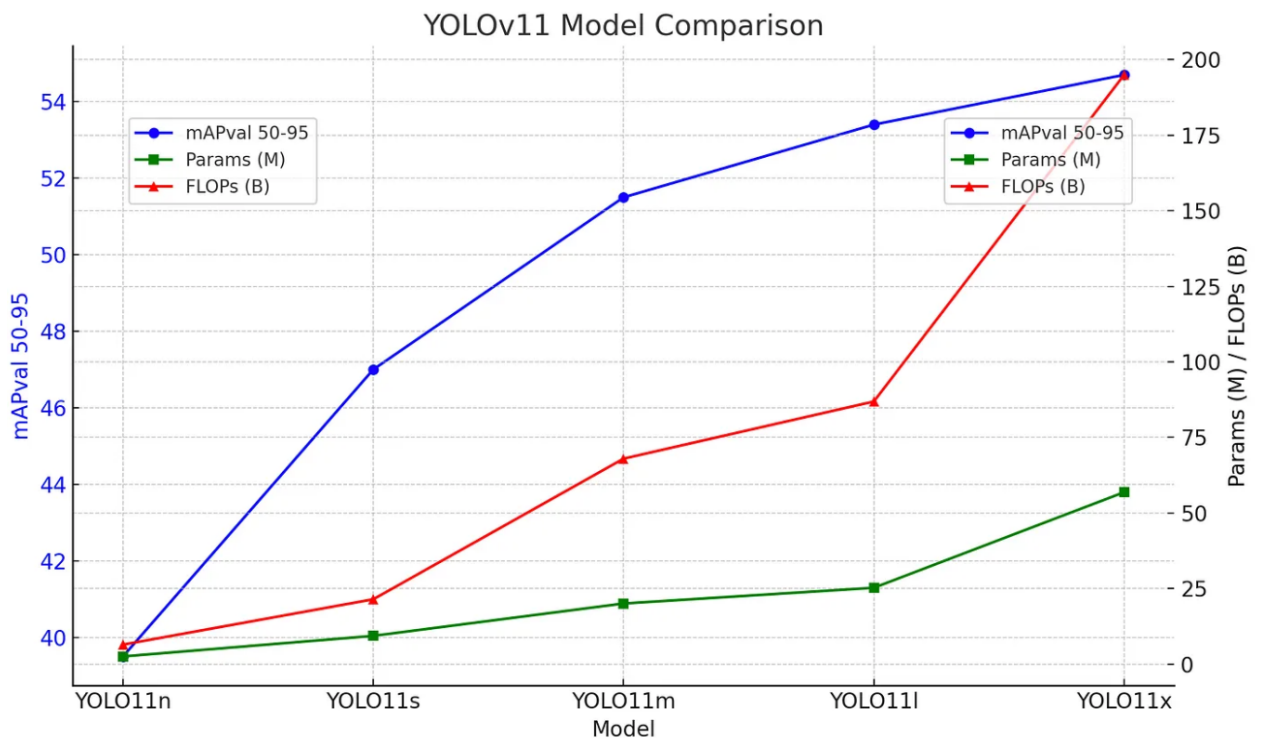
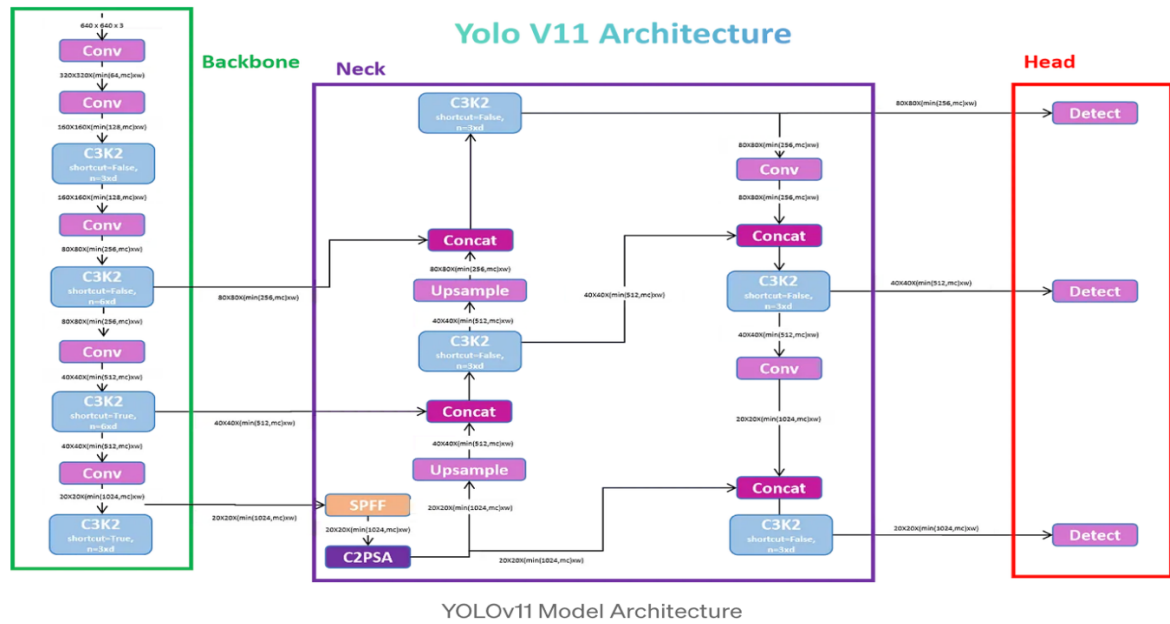
Output Directory.

- **Saved Weights:** runs/detect/train/weights/
- **Optimizer Removed:** Model weights (best.pt & last.pt) stripped to 40.5MB

Model Description.

- YOLOv11m is a state-of-the-art object detection model, fine-tuned on 1500+ annotated MRI scans of various tumor types. It achieved an impressive **mean Average Precision (mAP) of 94.2%**, with **precision of 92.7%** and **recall of 93.5%**, making it highly reliable for medical image analysis and tumor detection tasks.

10. Yolo11 Architecture.



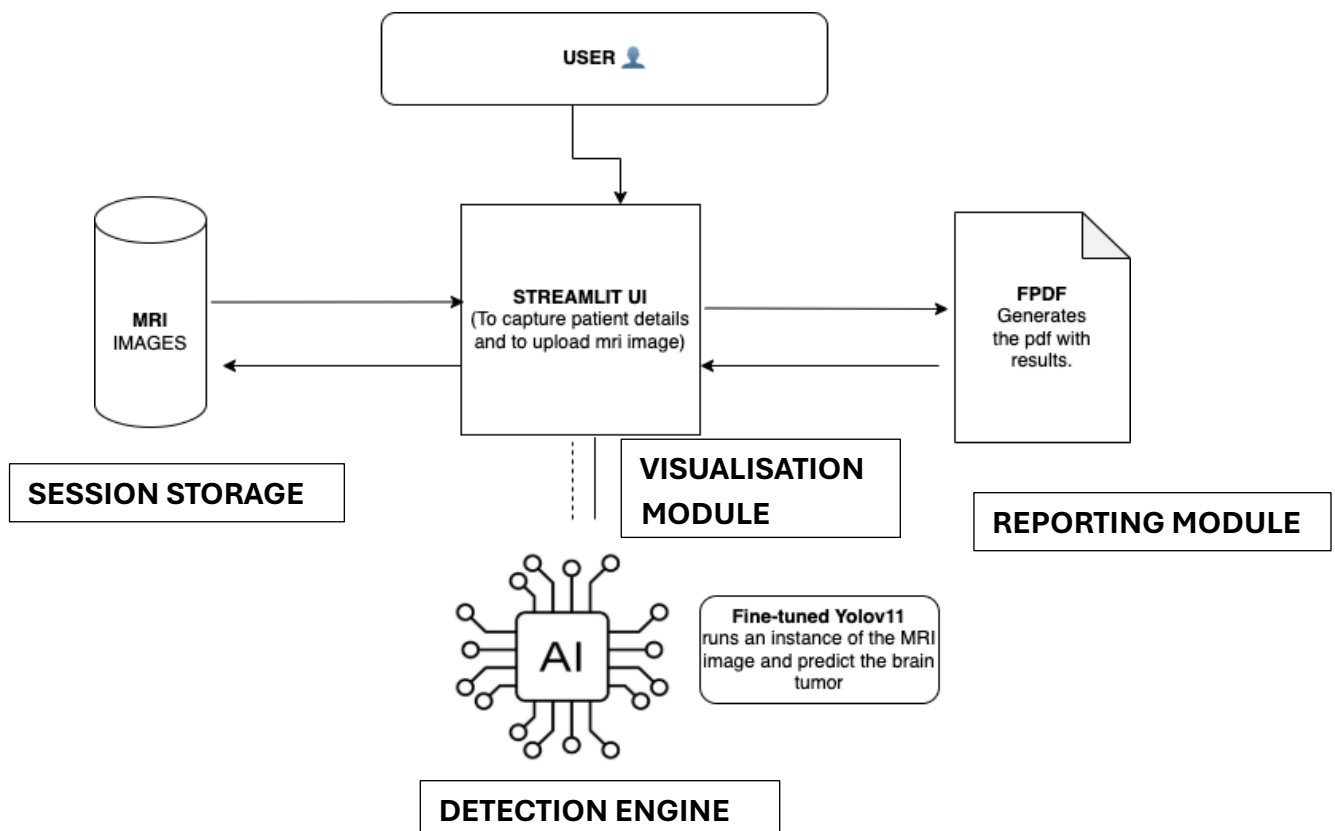
YOLOv11 Model comparison of different version for Detection Task

11. System Architecture.

The architecture is modular:

1. Data Input Module
2. Detection Engine
3. Visualization Module
4. Reporting Module

MRI scans are uploaded and preprocessed before detection. The model highlights tumors and generates reports.



12. User Guide.

1. Enter Patient ID, Patient Name, Gender and Age.

2. Upload an MRI scan image.

3. View analysis with bounding boxes.

4. Generate and download report.

13. Technical Implementation Details.

Image preprocessing includes resizing, normalization, and conversion to tensors. The detection engine performs inference, while Streamlit renders results. Reports are generated using FPDF.

14. UI Screenshots.

Share ★ ✎ ⋮

🧠 Brain Tumor Detection System

This application uses advanced AI technology to detect brain tumors from MRI scans. Upload your MRI image and get instant analysis with high accuracy.

Patient Information

Patient ID

Enter unique patient identifier

Patient Name

Enter full name

Gender

Male

Age

30

-

+

Upload MRI Scan

Upload MRI Scan

Drag and drop file here

Browse files

Detection Results

After uploading, click 'Detect Tumor' (in the left column) to analyze the scan.

Generate Medical Report

Generate PDF Report

Brain Tumor Detection System © 2025. All rights reserved.

Contact: leotalkofficial@gmail.com | Developed with ❤️ for healthcare professionals

Manage app

brain-tumor-vision.streamlit.app

Share ★ ✎ ⋮

Gender

Male

Age

30

-

+

Generate Medical Report

Generate PDF Report

Upload MRI Scan

Upload MRI Scan

Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files

About Brain Tumor Detection

This application uses fine-tuned YOLOv11, a state-of-the-art object detection model trained specifically for identifying brain tumors in MRI scans. The model has been fine-tuned on thousands of medical images to ensure high accuracy.

How to use:

1. Enter the patient's information in the form
2. Upload a clear MRI scan (JPEG, PNG format)
3. Click "Detect Tumor" to analyze the scan
4. Click "Generate PDF Report" to create your medical report
5. Download the PDF report using the button provided

Note: This tool is designed to assist medical professionals and should not replace professional medical diagnosis.

Brain Tumor Detection System © 2025. All rights reserved.

Contact: leotalkofficial@gmail.com | Developed with ❤️ for healthcare professionals

Manage app

15. Generated Output.

BRAIN TUMOR DETECTION REPORT

Contact: leotalkofficial@gmail.com

Generated on: February 27, 2025 17:53:36

PATIENT INFORMATION	
Patient ID:	ZW-1980-263
Patient Name:	Levi Ragnar
Gender:	Male
Age:	24

DETECTION RESULTS

TUMOR DETECTED

No.	Location	Confidence	Classification
1	(243, 318)	0.60	Benign
2	(179, 57)	0.57	Benign

MRI SCAN RESULTS



DISCLAIMER: This report is generated using an AI-based detection system and should be reviewed by a licensed medical professional. This tool is designed to aid in diagnosis, not replace professional medical judgment.

16. Conclusion.

This project demonstrates the power of deep learning in the medical domain. With high accuracy and real-time detection, it has the potential to assist radiologists in early tumor diagnosis.

17. Future Enhancements.

1. 3D MRI support.
2. Attention mechanisms.
3. Mobile version.
4. DICOM server integration.
5. Expert feedback tools.

18. References.

1. Ronneberger et al., U-Net: Biomedical Image Segmentation.
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4. Ultralytics YOLOv11 GitHub - <https://github.com/ultralytics/ultralytics>.
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